

Advanced Programming

Assignment 11 | Spring 2026

Name	Gaurav Nath
Roll No.	CSB24073
Subject	Advanced Programming
Assignment	Assignment No. 11
Semester	Spring 2026

Library System — OOP in Java

Question

Design a library system in Java with:

- A base/abstract class `LibraryItem` with common fields: title, year
- An abstract method `displayInfo()`
- Subclass `Book` — add author and genre
- Subclass `DVD` — add duration and genre

Implementation must clearly show:

- Use of abstraction — common structure defined in the base class
- Method overriding in each subclass
- Polymorphism using a collection of `LibraryItem` objects
- Additional feature: constructor overloading AND static item counter

Implementation

File 1: `LibraryItem.java` (Abstract Base Class)

```
// Abstract class - cannot be instantiated directly
public abstract class LibraryItem {

    private String title;
    private int    year;

    // Static counter - belongs to the class, shared across all objects
    private static int totalItems = 0;

    public LibraryItem(String title, int year) {
        this.title = title;
        this.year  = year;
        totalItems++; // increments every time any LibraryItem is created
    }
}
```

```

    }

    // Getters
    public String getTitle()          { return title; }
    public int    getYear()           { return year; }
    public static int getTotalItems() { return totalItems; }

    // Abstract method – every subclass MUST implement this or code won't
    compile
    public abstract void displayInfo();
}

```

File 2: Book.java (Subclass)

```

public class Book extends LibraryItem {

    private String author;
    private String genre;

    // Constructor 1 – no genre given, defaults to "General"
    public Book(String title, int year, String author) {
        this(title, year, author, "General");
    }

    // Constructor 2 – full constructor
    public Book(String title, int year, String author, String genre) {
        super(title, year); // calls LibraryItem constructor
        this.author = author;
        this.genre  = genre;
    }

    public String getAuthor() { return author; }
    public String getGenre()  { return genre; }

    @Override
    public void displayInfo() {
        System.out.println("-----");
        System.out.println("Type    : Book");
        System.out.println("Title   : " + getTitle());
        System.out.println("Author  : " + author);
        System.out.println("Genre   : " + genre);
        System.out.println("Year    : " + getYear());
        System.out.println("-----");
    }
}

```

File 3: DVD.java (Subclass)

```

public class DVD extends LibraryItem {

    private int    duration;
    private String genre;

    // Constructor 1 – no genre given, defaults to "Uncategorized"
    public DVD(String title, int year, int duration) {
        this(title, year, duration, "Uncategorized");
    }

    // Constructor 2 – full constructor

```

```

    public DVD(String title, int year, int duration, String genre) {
        super(title, year); // calls LibraryItem constructor
        this.duration = duration;
        this.genre = genre;
    }

    public int getDuration() { return duration; }
    public String getGenre() { return genre; }

    @Override
    public void displayInfo() {
        System.out.println("-----");
        System.out.println("Type : DVD");
        System.out.println("Title : " + getTitle());
        System.out.println("Genre : " + genre);
        System.out.println("Duration : " + duration + " mins");
        System.out.println("Year : " + getYear());
        System.out.println("-----");
    }
}

```

File 4: Main.java (Driver — Polymorphism Demo)

```

import java.util.ArrayList;
import java.util.List;

public class Main {
    public static void main(String[] args) {

        // Constructor overloading - with and without genre
        Book b1 = new Book("The Alchemist", 1988, "Paulo Coelho", "Fiction");
        Book b2 = new Book("Clean Code", 2008, "Robert C. Martin");
        DVD d1 = new DVD("Inception", 2010, 148, "Sci-Fi");
        DVD d2 = new DVD("Planet Earth", 2006, 200);

        // POLYMORPHISM - LibraryItem reference holds Book and DVD objects
        List<LibraryItem> library = new ArrayList<>();
        library.add(b1);
        library.add(b2);
        library.add(d1);
        library.add(d2);

        System.out.println("===== LIBRARY CATALOGUE =====");
        for (LibraryItem item : library) {
            item.displayInfo(); // Java picks correct version at runtime
        }

        // Static counter - called on the class, not on any object
        System.out.println("Total items in library: " +
            LibraryItem.getTotalItems());
    }
}

```

OOP Concepts Demonstrated

Concept	Where	How
Abstraction	LibraryItem	abstract class + abstract method forces subclass to implement displayInfo()
Constructor Overloading	Book, DVD	Two constructors: with/without genre, chained via this(...)
Inheritance	Book, DVD	Both extend LibraryItem and inherit title, year, and counter
Method Overriding	@Override displayInfo()	Each subclass provides its own display logic for the same method name
Polymorphism	List<LibraryItem>	Loop calls displayInfo() — Java picks correct subclass version at runtime
Static Counter	totalItems in LibraryItem	Class-level field; incremented in constructor; tracks all items created

Compile & Run

```
cd LibrarySystem
javac LibraryItem.java Book.java DVD.java Main.java
java Main
```

Output Screenshots

Terminal output:

```

PS D:\Downloads D Folder\Advanced Programming 2> & 'C:\Program Files\Java\jdk-26.0.1\bin\java.exe' '--enable-preview' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\gaura\AppData\Roaming\Code\User\workspaceStorage\9f823240be34e422c5f6eea18358f704\redhat.java\jdt_ws\Advanced Programming 2_6ff7544\bin' 'Main'
===== LIBRARY CATALOGUE =====
-----
Type   : Book
Title  : The Alchemist
Author : Paulo Coelho
Genre  : Fiction
Year   : 1988
-----
Type   : Book
Title  : Clean Code
Author : Robert C. Martin
Genre  : General
Year   : 2008
-----
Type   : DVD
Title  : Inception
Genre  : Sci-Fi
Duration : 148 mins
Year   : 2010
-----
Type   : DVD
Title  : Planet Earth
Genre  : Uncategorized
Duration : 200 mins
Year   : 2006
-----
Total items in library: 4
PS D:\Downloads D Folder\Advanced Programming 2>

```

VS Code file structure:

